

# How error-driven learning shapes distributional learning of a familiar phonetic category

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## The Problem

Phonetic categories are learned through distributional learning (Maye, Werker, & Gerken, 2002). One category if unimodal, two if bimodal

- ▶ Research suggests that this is based on prediction and prediction error (Olejarczuk, Kapatsinski, & Baayen, 2018)
- ▶ In what way each token contributes to learning a **novel** representation of a **known** category is unclear
- ▶ **Goal:** Investigate if distributional learning of a known category is statistical (Hebb, 1949; Saffran, 2020) or **error-driven** (Olejarczuk, Kapatsinski, & Baayen, 2018; Rescorla & Wagner, 1972)

## Research Question

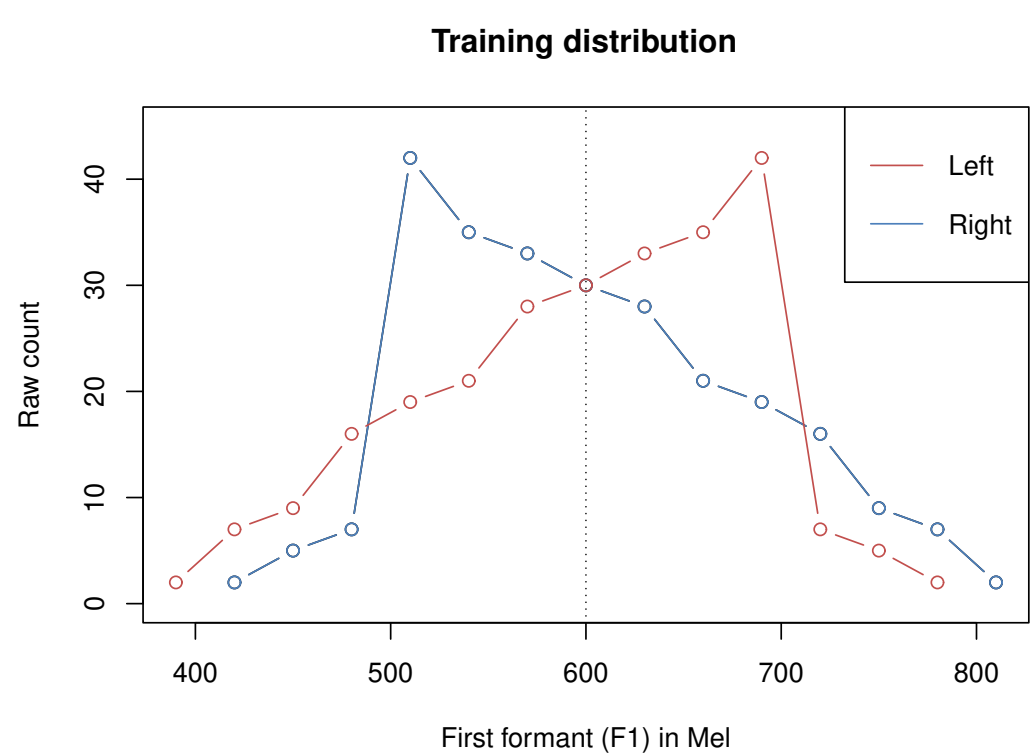
Do less frequent exemplars have a stronger effect on the phonetic category representation of /ɛ:/?

- ▶ Train two groups – each with a different distribution of **F1 values** (different mode and skew, but same mean)
- ▶ Compare **category goodness ratings** and **productions** (before/after training)
- ▶ If distributional learning is error-driven, representation will shift towards the distributions’ **tails**

## The Method

- ▶ Two different learner groups: One had the right skew, another the **left skew** distribution
- ▶ Both received the same pretest and posttest (Category Goodness rating using a 7 point Likert scale)
- ▶ F1 values manipulated in **30 Mel** increments

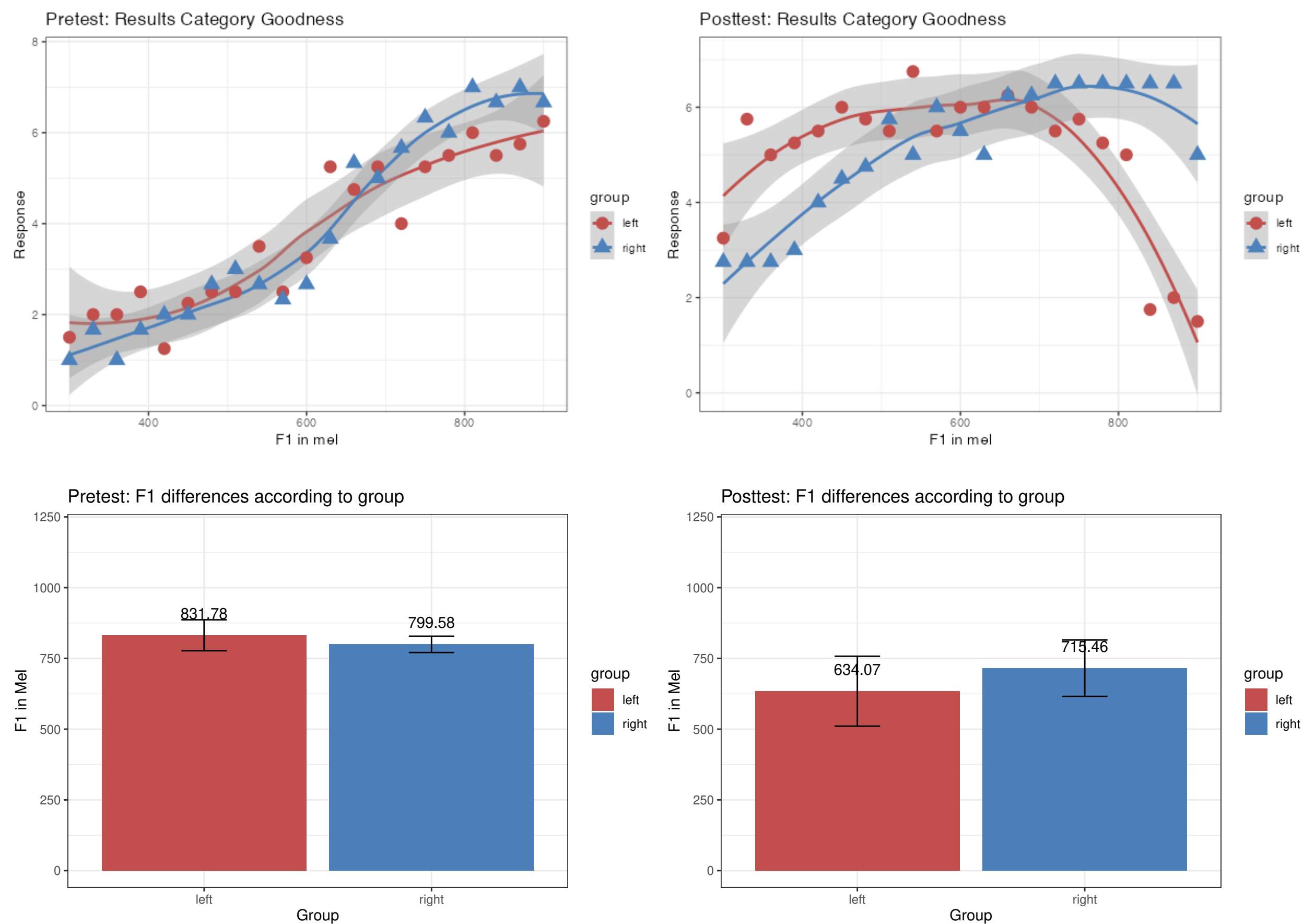
## The Stimuli



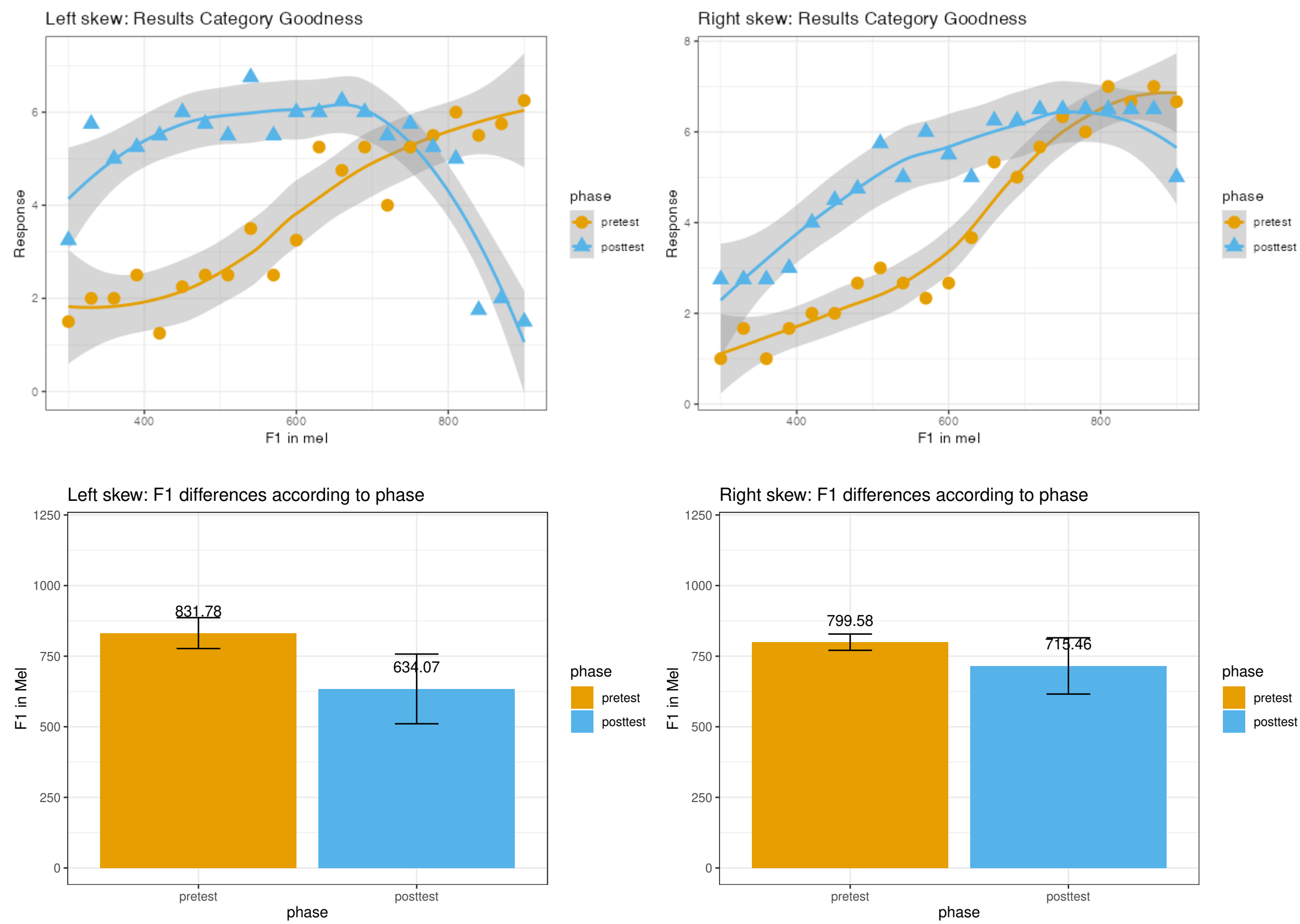
- ▶ Mean values for both = 600 Mel
- ▶ Mode Right skew = 510 Mel
- ▶ Mode **Left skew** = 690 Mel

|                  | Range       | Count |
|------------------|-------------|-------|
| Pretest          | 300-900 Mel | 21    |
| Right skew       | 420-810 Mel | 256   |
| <b>Left skew</b> | 390-780 Mel | 256   |
| Posttest         | 300-900 Mel | 21    |

## Results



- ▶ Curve for **left skew** has better ratings for lower frequencies (lower F1 better, higher F1 worse)
- ▶ Curve for right skew has better ratings for higher frequencies (lower F1 worse, higher better)



- ▶ Pretest vs. posttest differences within each group were significant
- ▶ One exception though: No significant difference for production in right skew

## Discussion

- ▶ Categorical goodness judgments: Significant difference between **pretest** and **posttest** for **both** right skew and **left skew** groups (but no difference between groups)
- ▶ Production: Significant difference between **pretest** and **posttest** for **left skew** group **only**
- ▶ All other comparisons did not differ significantly
- ▶ Similar trend as in Olejarczuk, Kapatsinski, & Baayen (2018) – in line with **error-driven learning**
- ▶ Analysis is based on only 8 participants though (168 data points for category goodness; 42 data points for production)

## References

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